$f = o(g) \rightarrow f(n) < g(n)$

$f = \Omega(g) \rightarrow f(n) > g(n)$

1)

a) $f(n) = 2n+300$, $g(n) = 100n+2$

$f(n) = \Theta(g(n))$, $f = o(g)$ and $g = O(f)$ $(f = \Omega(g))$

$f(n) = O(g(n)) \Rightarrow 2n+300 = \Theta(100n+2)$

b) $f(n) = n^{3/4}$ $g(n) = n^{2/3}$

$f(n) = \Omega(g(n))$, $n^{3/4} > n^{2/3}$

c) $f(n) = 10n^2 + (\log n)^3$ $g(n) = n + \log n$

$f(n) = \Theta(g(n))$

$10n^2 + (\log n)^3 = \Theta(n + \log n)$

d) $37n \log(10^6 n) = f(n)$ $g(n) = n \log n$

$f(n) = \Theta(g(n)) \rightarrow 37n \log(10^6 n) = \Theta(n \log n)$

e) $f(n) = n^2 \log n$ $g(n) = n^{2.0001} (\log n)^2$

$f(n) = \Theta(g(n))$

$\Rightarrow n^2 \log n = \Theta(n^{2.0001} (\log n)^2$

f) $f(n) = (\log n)^2$ $g(n) = n/\log n$

$f(n) = o(g) \rightarrow (\log n)^2 = o(n/\log n)$

$\sum_{i=1} i^2 \rightarrow n^3 \geq \sum_{i=1}^{n} i^2$

$n \geq 1$

g) $f(n) = n/\log n \qquad g(n) = (\log n)^{\log n}$

$f(n) = o(g(n)) \rightarrow n/\log n = o(\log n)^{\log n}$

h) $f(n) = (\log n)^3 \quad g(n) = \sqrt[3]{n}$

$f(n) = o(g(n)) \rightarrow (\log n)^3 = o(\sqrt[3]{n})$

i) $f(n) = 3^{\log_2 n} \quad g(n) = n^{\log_2 3}$

$f(n) = o(g(n)) \rightarrow 3^{\log_2 n} = n^{\log_2 3}$

j) $f(n) = 3^n \qquad g(n) = n^4 2^n$

$f(n) = \Omega(g(n))$

$3^n = \Omega(n^4 2^n)$

k) $f(n) = n! \qquad g(n) = 2^n$

$f(n) = \Omega(g(n))$

$n! = \Omega(2^n)$

l) $f(n) = 2(\log n)^2 \qquad g(n) = (\log n)^{\log n}$

$f(n) = O(g(n)) \qquad 2^{(\log n)^2} = O(\log n)^{\log n}$

2)

a) $\sum_{i=1}^{n} i^2$ Given / $\sum_{i=1}^{n} i^2 \Rightarrow \sum_{i=1}^{n} n^2$

$i \le n, \rightarrow \boxed{\sum_{i=1}^{n} i^2 \le \sum_{i=1}^{n} n^2}$

b)

$n^2 \sum_{i=1}^{n} 1 \ge \sum_{i=1}^{n} i^2 \rightarrow n^2(n) \ge \sum_{i=1}^{n} i^2 \rightarrow n^3 \ge \sum_{i=1}^{n} i^2$

n^2 is constant

c) $\sum_{i=1}^{n} i^2 = O(n^3)$

$\hookrightarrow 0 \le \sum_{i=1}^{n} i^2 \le n^3$ when $n \ge 1$

$\therefore \sum_{i=1}^{n} i^2 = O(n^3)$

g) $f(n) =$

$f(n)$

h) $f(n) =$

$f($

i) $f(n)$

j) $f($

k) f

l) $f($

3

(a) $5n^2 \log_n + 3n^2 + 7n \log_n + 9n + 2$

If $f(n) \leq t(n)$, $t(n) \geq f(n)$

$5n^2 \log_n + 3n^2 + 7n\log_n + 9n + 2 \leq cn^2 \log_n$ → c is constant

Thus, $\Rightarrow 5n^2 \log_n + 3n^2 + 7n \log_n + 9n + 2 = O(n^2 \log_n)$

b) $f(n) = 3n^2(\sqrt{(n)}) + 5n\log_n + 7$

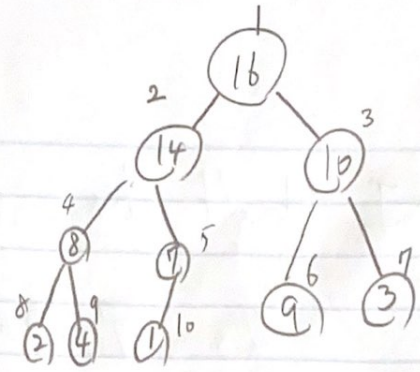
$3n^2(\sqrt{n}) + 5n\log_n + 7 \leq cn^2\sqrt{n}$ c → constant

therefore, $3n^2(\sqrt{n}) + 5n\log_n + 7 = O(n^2\sqrt{n})$

4)
a) I don't know how to solve this problem

3
(a)

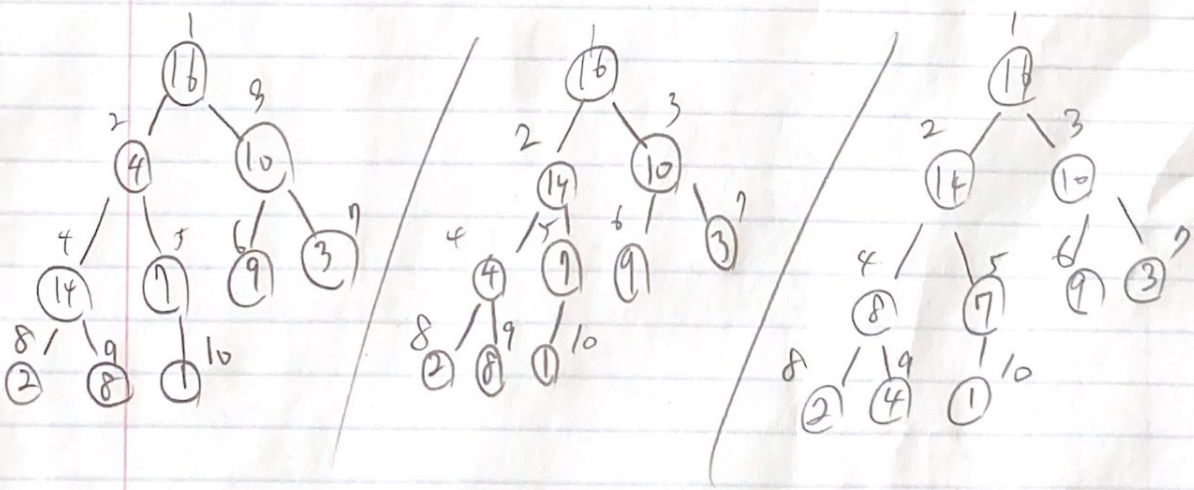
5 pesudocode




```
ParentNode (j)
  return j/2;
MoveRight (j)
  return 2j+1;
MoveLeft (j)
  return 2j;
```

Running time → $O(r)$

$T(n) = T(2n/3) + 1$

$T(n) = O(\log n)$

⇒ characterize the running time on a height of node → $O(r)$